

FRACTURES — LATE LOCAL COMPLICATIONS (Revision Table)

Bold = high-yield. Perkins' timetable (normal): callus 2-3 wks (both); UNION upper 4-6 / lower 8-12 wks; CONSOLIDATION upper 6-8 / lower 12-16 wks.

BONE UNION PROBLEMS

Complication	Cause / pathophysiology	Clinical & X-ray	Treatment
Delayed union	Fracture unites slower than usual. Inadequate blood supply, severe soft-tissue damage, periosteal stripping, excessive traction, insufficient splintage, infection, intact fellow bone	Fracture tenderness (esp. on stress). X-ray: visible fracture line, little callus / periosteal reaction	Conservative: eliminate cause, immobilize, exercise. Operative if >6 months + no callus → internal fixation + bone grafting
Non-union	Will NEVER unite without intervention ; gap filled by fibrous tissue (pseudoarthrosis). Improper Tx of delayed union, too large gap, soft-tissue interposition (periosteum/muscle/cartilage)	Painless movement at fracture site. X-ray: clear fracture, ends rounded/smooth/sclerotic. Atrophic = inactive/avascular/tapered; Hypertrophic = excess bone but can't bridge gap	Hypertrophic → rigid fixation (± graft). Atrophic → fixation + bone grafting
Malunion	Unite in unsatisfactory position (angulation/rotation/shortening). Failure to reduce or hold reduction; collapse of comminuted/osteoporotic bone	Deformity, shortening, limited movement	Angulation >15° (long bone) → osteotomy + IF. Marked rotation → osteotomy + IF. Shortening >3cm (lower limb) → raised boot or bone op

JOINT & SOFT-TISSUE COMPLICATIONS

Complication	Cause / pathophysiology	Clinical & X-ray	Treatment
Joint stiffness	Common after immobilization (knee, elbow, shoulder, hand). Oedema/fibrosis of capsule/ligaments, intra/peri-articular adhesions, synovial adhesions from haemarthrosis	Restricted joint movement	Prevent: exercise + correct splint position. Physio; intra-articular adhesions → manipulation under anaesthesia + CPM ; contracted tissue → surgical release
Myositis ossificans	Heterotopic ossification in muscle after injury. Elbow dislocation , blow to brachialis/deltoid/quadriiceps. Also unconscious/paraplegic (no local injury)	Pain, tenderness, swelling, stiffness. Extreme → bone bridges joint (extra-articular ankylosis). X-ray: normal early → fluffy calcification	Rest early → gentle active movement → excise bony mass once stabilized. Anti-inflammatories ↓ stiffness
Avascular necrosis (AVN)	Circumscribed necrosis — interrupted arterial flow / slowed venous outflow. Sites: femoral head, femoral condyles, humeral head, capitulum, scaphoid (proximal), talus (body), lunate. A/w sickle cell, steroids, alcohol, caisson, Gaucher's, Perthes'	Joint pain, stiffness, swelling, restricted movement. X-ray: ↑bone density, subarticular fracturing, deformity	Avoid weight-bearing, revascularization (vascularized bone graft) , excise avascular segment, prosthetic replacement
Algodystrophy (CRPS)	= Sudeck's atrophy / reflex sympathetic dystrophy. Post-trauma, usually foot/hand after trivial injury	Continuous burning pain ; early: swelling, redness, warmth; later: skin/muscle atrophy, grossly restricted movement. X-ray: patchy osteoporosis (rarefaction)	Physiotherapy (elevation + active exercise), anti-inflammatories, sympathetic block / sympatholytic (guanethidine)
Post-traumatic OA	Direct: joint fracture + damaged cartilage → OA within months . Secondary: healed but irregular surface → stress → OA years later	Pain, stiffness, swelling, deformity, restricted movement	Analgesics/anti-inflammatory, physio, weight reduction, realignment osteotomy (young), arthroplasty (>60y)

OTHER LISTED COMPLICATIONS (mentioned, not detailed)

Group	Items
Also includes	Joint instability · Muscle contracture (Volkmann's ischaemic contracture) · Tendon lesions (e.g. rupture EPL after Colles') · Nerve compression / entrapment · Growth disturbance (physeal injury in children) · Bed sores (from immobility)